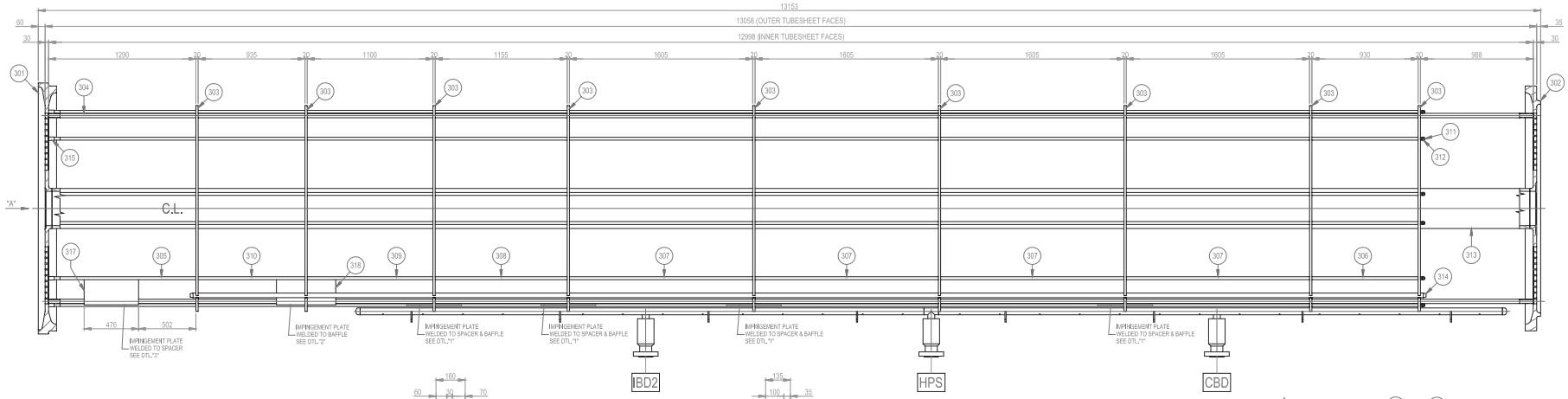
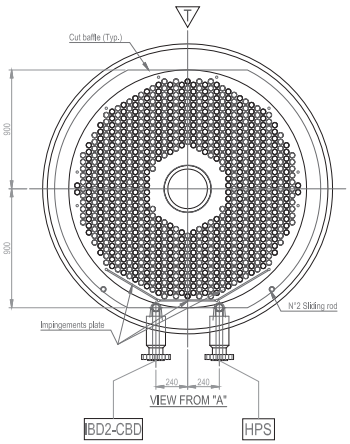
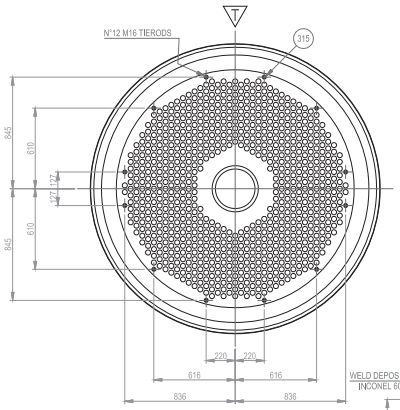
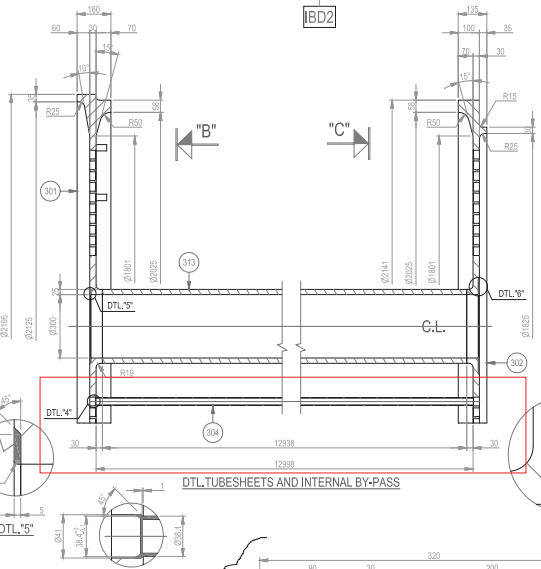
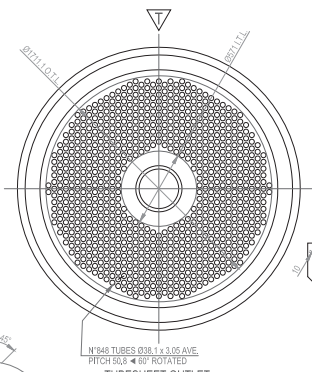
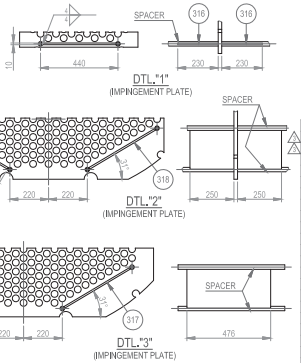


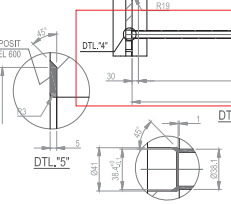


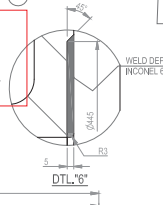


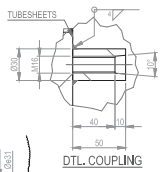


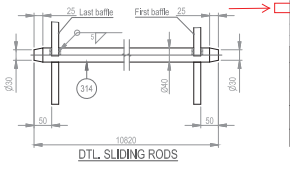










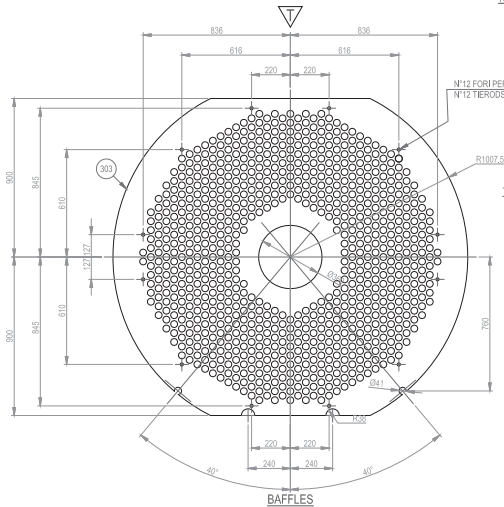


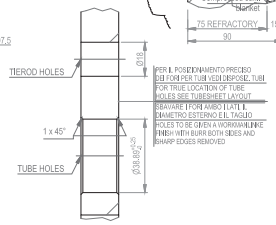
POS.	DESCRIZIONE/DESCRIPTION	DIMENSIONI/DIMENSIONS	MATERIALE/MATERIAL	Q.TY	CERTIFICATO CERTIFIED	SPARE PARTS	SPARE FOR MAINTENANCE	NOTES
326	Ferula/Ferule	See dwg.	ALLOY 801H	848	3.1			
325	Piastrina ferula/Ring for ferule	Ø447 x Ø31 x Thk.3	ALLOY 801H	848	3.1			
324	Nervatura tubo/Pipe stiffening	95 x 95 x Thk.10	SA-516 Gr.70N	16	3.1			
323	Tee normale/Equal Tee	Ø2" x Sch.XS	SA-234 WPB	3	3.1			
322	Fondello/Caps	Ø2" x Sch.XS	SA-234 WPB	4	3.1			
321	Tubo per distrib. Int./Pipe for Int. distrib.	Ø2" x Sch.XS Lg.4872	SA-106 Gr.B	1	3.1			
320	Tubo per distrib. Int./Pipe for Int. distrib.	Ø2" x Sch.XS Lg.2436	SA-106 Gr.B	2	3.1			
319	Tubo per distrib. Int./Pipe for Int. distrib.	Ø2" x Sch.XS Lg.4556	SA-106 Gr.B	2	3.1			
318	Flanti rompiflusso/Impingment plate	250 x 435 x Thk.10	SA-516 Gr.70N	4	3.1			
317	Flanti rompiflusso/Impingment plate	476 x 435 x Thk.10	SA-516 Gr.70N	2	3.1			
316	Flanti rompiflusso/Impingment plate	230 x 414 x Thk.10	SA-516 Gr.70N	8	3.1			
315	Manicotto/Coupling	Ø630 x Lg.50	SA-182 F22 CL3	12	3.1			
314	Tondo di scorrimento/Sliding rod	Ø640 x Lg.10820	SA-36	2	2.2			
313	Tubo per By-Pass/Pipe for By-Pass	Ø6350 x Thk.25 Lg.12938	SA-533 Gr.B Cl.2	1	3.1			
312	Dado/Nut	M16	SA-36	24	2.2			
311	Tirand fissi/Tierods	M16 x Lg.12050	SA-36	12	2.2			
310	Tubi distanziali/Spacer tubes	Ø635,4 x Thk.2,11 Lg.935	SA-179	12	2.2			
309	Tubi distanziali/Spacer tubes	Ø635,4 x Thk.2,11 Lg.1100	SA-179	12	2.2			
308	Tubi distanziali/Spacer tubes	Ø635,4 x Thk.2,11 Lg.1155	SA-179	12	2.2			
307	Tubi distanziali/Spacer tubes	Ø635,4 x Thk.2,11 Lg.1605	SA-179	48	2.2			
306	Tubi distanziali/Spacer tubes	Ø635,4 x Thk.2,11 Lg.930	SA-179	12	2.2			
305	Tubi distanziali/Spacer tubes	Ø635,4 x Thk.2,11 Lg.1240	SA-179	12	2.2			
304	Tubi di scambio/Tube exchanger	Ø38,1 x Thk.3,05 (ave) Lg.13000	SA-213 Gr.T11	848	3.1			
303	Diaframmi/Baffles	Ø2015 x Thk.20	SA-516 Gr.70N	9	2.2			
302	Piastra tubiera uscita/Tubesheet outlet	Ø6741 x Thk.135	SA-508 22 CL3	1	3.1			
301	Piastra tubiera entrata/Tubesheet inlet	Ø6785 x Thk.160	SA-508 22 CL3	1	3.1			

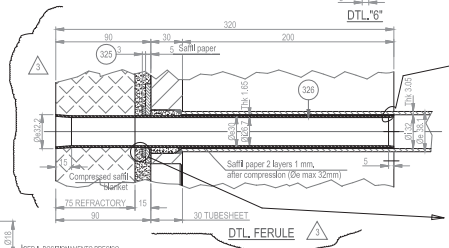
Materiale soggetto a controllo chimico. Material tested by chemical analysis.
Materiale con certificato tipo produttore. Material with manufacturer certificate.
Materiale senza certificato ad produttore. Material without manufacturer certificate.
3,1
2,2
MATERIALS TEST SYMBOL (Certification Acc. TO EN-10204) Simbolismo collaudo materiali

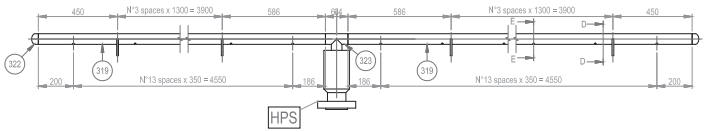
- Il materiale elencato in distinta è da intendersi valido per N°1 apparecchio.
Quantity indicated on material list are for one shell.

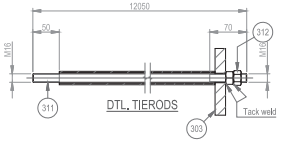
LISTA MATERIALI / MATERIALS LIST

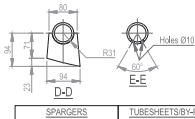


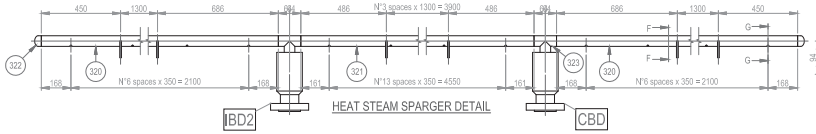


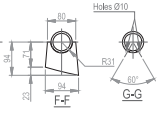


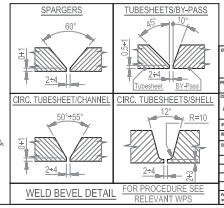












REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
3	15/05/2014	Revised as Indicated	Pisanti	Sarti	Sarti
2	25/10/2013	Revised as Indicated	Mazzolani	Sarti	Sarti
1	25/10/2013	Revised as Indicated	Mazzolani	Sarti	Sarti
0	29/04/2013	Issued for approval	Mazzolani	Sarti	Sarti

SUISIO - BERGAMO - ITALY

CLIENTE / CUSTOMER: Fimco
ORDINE CLIENTE / PURCHASE ORDER: FZMC-13450-R1
IMPIANTO E LOC. / LOCATION & UNIT: Ammonia Urea Project
SERVIZIO / SERVICES: Sec. Reformer Effluent Waste Heat Boiler
TITOLO / TITLE: Tube bundle drawing

SCALA/FORMATO SCALE/FORMAT	NO / A1	TIPO / TEMA / TYPE	NUMERO APPAR. / NUMBER REQUIRED	SIGLA / ITEM No.
DATA / DATE	29/04/2013	BXM SPEC.	1	3-E-1401
DISEGNATO / DRAWN	Mazzolani			
CONTROLL. / CHECKED	Passerini			
APPROVATO / APPROVED	Sarti			

DIMENSIONI / SIZE	NUM. DI FABBRICA / SERIAL NUMBER	ANNO DI COS. / YEAR OF CONS.	COMM. / JOB	NUM. DIS. / DWG. NUM.	REV.
13000 x 2025	12542	2014	7523	7523-01-300-01	3

DISEGNO DEFINITIVO
Il cliente ha già preso visione del disegno ed ha approvato le sedi non evidenziate dall'ultima rev.
VALID FOR CONSTRUCTION
Drawing already approved by client except where indicated with last revision mark.
DATA: 15/05/2014 FIRMA: Sarti

REV.	ISSUE DATE	DESCRIPTION	DISCIPLINE
3	15/05/14	REVISED AS INDICATED	SP SS SS
2	25/10/13	REVISED AS INDICATED	PM FG SS
1	31/05/13	REVISED AS INDICATED	PM FG SS
0	29/04/13	ISSUED FOR APPROVAL	PM FG SS

OWNER: LORDEGAN PETROCHEMICAL COMPANY
Project: LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR: HAMPA ENGINEERING CORPORATION
DETAIL ENGINEER: MEDDO Engineering & Consulting
TITLE: TUBE BUNDLE DRAWING
Project No.2103 | HAMPA | Doc. No.: 7523-01-300-01 | REV. 3
Scale: | Vendor | Sheet: of | 2103-A-VD-255-A1-MD-025

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